

BAYKAR TEKNOLOJİ İNSAN KAYNAKLARI DİREKTÖRLÜĞÜ'NE**İlgili Pozisyon: RF ve Mikrodalga Sistemleri Geliştirme (2027 Bahar Dönemi Stajı)**

İstanbul, Türkiye

Aday: Efe Demirer

Elektrik-Elektronik Mühendisliği (GNO: 3.73 / 4.00)

Yaşar Üniversitesi (%100 ÖSYM Tam Burslu)

Sayın Yetkili,

Milli Teknoloji Hamlesi vizyonu ile Bayraktar TB2, TB3, AKINCI ve KIZILELMA gibi insansız hava araçları ekosistemimize küresel liderlik kazandıran **BAYKAR Teknoloji** bünyesinde; **RF ve Mikrodalga Sistemleri Geliştirme** biriminde stajyer mühendis olarak görev almak amacıyla başvurumu sunmaktayım.

Yaşar Üniversitesi Elektrik-Elektronik Mühendisliği 4. sınıf öğrencisiyim (%100 Burslu, 3.73 GNO, Yüksek Onur Öğrencisi). Akademik eğitimimde bu dönem almakta olduğum **EEE 4340 Antenler ve Propagasyon**, Elektromanyetik Alan Teorisi ve Mikrodalga Ölçüm disiplinlerinde edindiğim teorik temeli, doğrudan savunma sanayii Ar-Ge ve test laboratuvarlarında somut projelere dönüştürdüm.

ASELSAN REHİS Test Merkezleri Müdürlüğü bünyesindeki stajımda; yankısız odalarda (anechoic chamber) anten ışıma deseni yakın alan ölçümleri için **Kartezyen X-Y tarama otomasyonu ve C# GUI** kontrol yazılımı geliştirdim. Ayrıca Vektör Ağ Analizörü (VNA) tam 2-port kalibrasyonları, ölçüm referans düzleminin kablo uçlarına taşınması, RF bileşen karakterizasyonu ve **amfilerin P1dB sıkıştırma (compression) testlerinde** bilfiil görev aldım. RF soğurucu malzemelerin performansını NRL Arch Test yöntemiyle karakterize ettim ve **Gerçek Zamanlı Spektrum Analizörü (Real-Time Spectrum Analyzer)** için Ethernet TCP üzerinden C/C++ API entegrasyonu ve anlık tepe noktası (peak search) algoritması tasarladım.

2027 Bahar Dönemi Staj Takvimi Kapsamında (01.02.2027 – 22.05.2027): Üniversite 4. sınıf ders programımı koordine ederek, staj süresince **haftada en az 3 gün (veya birimin ihtiyacına göre daha fazla)** düzenli, kesintisiz ve tam mesaiyle Baykar Ar-Ge tesislerinde aktif görev almaya hazırım.

Baykar'ın milli İHA platformlarında görev yapan anten, RF veri bağı (data link), radar ve mikrodalga alt sistemlerinin tasarımı, simülasyon ve test süreçlerine yüksek sorumluluk bilinciyle katkı sağlamayı hedefliyorum. Çalışmalarımı özetleyen teknik portfolyom ekte sunulmuş olup, detaylı projelerime [efedemirerr.github.io](https://github.com/efedemirerr) adresinden de erişilebilir. Başvurumu değerlendirmeye aldığınız için teşekkür eder, saygılarımı sunarım.

Efe DEMİRER

Aday RF ve Mikrodalga Sistemleri Mühendisi · Yaşar Üniversitesi EEM | İzmir, Bornova | +90 554 638 86 92 | efedemirerr.tr@gmail.com



Efe Demirer

Candidate RF & Microwave Systems Engineer · Antennas & Test Automation

Senior Electrical and Electronics Engineering student at Yaşar University specializing in RF & Microwave systems, antenna radiation pattern testing, and high-frequency instrumentation automation. Laboratory and defense experience at ASELSAN REHİS Test Centers Directorate conducting planar near-field antenna scanning, VNA 2-port calibrations, RF amplifier characterization, and real-time spectrum analysis.

[Live Portfolio \(GitHub Pages\)](#)[LinkedIn Profile](#)efedemirer.tr@gmail.com[GitHub: efedemirerr](#)

SKILLS & HIGHLIGHTS

Core RF & Microwave Highlights · ASELSAN REHİS

[ASELSAN REHİS RF Intern](#)[Antennas & Near-Field Scanning](#)[VNA & Spectrum Automation](#)[P1dB Amplifier Compression](#)[Antennas & Propagation](#)[VNA 2-Port Calibration](#)[Near-Field Anechoic Chambers](#)[RF Power Amplifiers \(P1dB\)](#)[NRL Arch Testing \(Absorbers\)](#)[C# .NET Test Automation](#)

RF & Microwave Engineering

[Antennas & Propagation](#)[Vector Network Analyzers \(VNA\)](#)[Real-Time Spectrum Analyzers](#)[VNA Full 2-Port Calibration](#)[P1dB Compression Testing](#)[RF Component Characterization](#)[Near-Field Anechoic Testing](#)[NRL Arch Absorber Testing](#)[S-Parameters & Impedance Matching](#)

RF Test Automation & Software

[C#](#)[.NET](#)[Windows Forms](#)[Python](#)[API Integration](#)[SCPI & G-Code](#)[Serial \(RS-232\)](#)[Ethernet TCP/IP](#)[MATLAB](#)

Motion & Scanning Hardware

[Cartesian X-Y Scanners](#)[Stepper Motor Control](#)[CNC Shield V3](#)[GRBL Firmware](#)[Limit Switch Safety Interlocks](#)[Arduino & STM32](#)

Electrical & Energy Systems

[Solar PV Design \(GES\)](#)[PVsyst Simulation](#)[ETAP Power Analysis](#)[AutoCAD](#)[Reactive Power Compensation](#)[High Voltage Inspection](#)[RCD & Grounding Tests](#)

Simulation & Tools

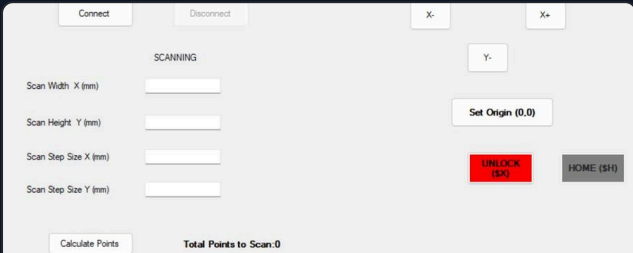
[Simulink](#)[LTspice](#)[Proteus](#)[Git / GitHub](#)[Visual Studio](#)[Microsoft Project](#)

FEATURED RF & MICROWAVE DEFENSE PROJECTS

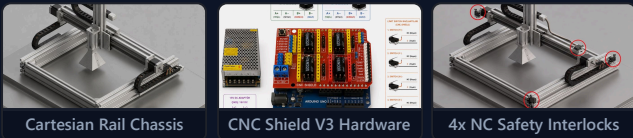
ASELSAN REHİS · ANTENNA TEST AUTOMATION

Automated CNC Antenna Scanning Test System & Control GUI

Anechoic Chamber Planar Near-Field Testing · 2026



EFEScan C# .NET Control GUI · Live Scan Grid Coordinate Tracking



Cartesian Rail Chassis

CNC Shield V3 Hardware

4x NC Safety Interlocks

Motion: Cartesian X-Y (GRBL)

Safety: 4x NC Limit Switches

Interface: Serial COM (115200)

Application: Near-Field Chamber

Autonomous Cartesian X-Y planar antenna scanning system engineered for ASELSAN REHİS Test Centers Directorate, replacing manual probe positioning with high-precision automated radiation pattern measurements in anechoic chambers.

- ▶ **Hardware & Motion Control:** Interfaced PC with Arduino Uno + CNC Shield V3 running GRBL firmware over asynchronous Serial Port (115200 Baud).
- ▶ **G-Code Protocol:** Implemented real-time G-Code stream generation (G90/G91/G92/G1) with centered grid coordinate pathway calculations and dynamic step resolution.
- ▶ **Hardware Safety Interlocks:** Integrated 4 mechanical Normally-Closed (NC) limit switches for hardware-level collision prevention, triggering instant GRBL ALARM mode and motor cutoff.
- ▶ **Multi-Threaded Execution:** Designed asynchronous BackgroundWorker threads eliminating UI freezing, paired with live bitmap scan visualization and feedrate-calibrated motion delays.

C# .NET

GRBL G-Code

Arduino & CNC Shield

Hardware Limit Switches

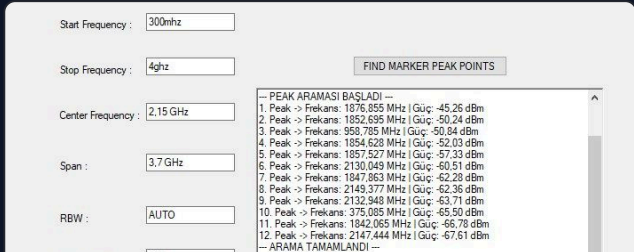
Near-Field Scanning

Anechoic Chamber

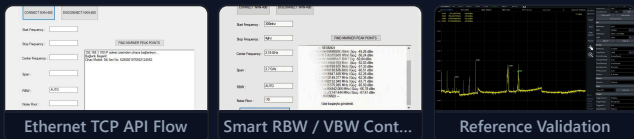
ASELSAN REHİS · RF SIGNAL ANALYSIS & INSTRUMENTATION

Real-Time Spectrum Analyzer API & Custom GUI

High-Speed Ethernet API & Real-Time Peak Detection · 2026



Custom C# GUI Real-Time Spectrum Trace & Active Peak Detection



Ethernet TCP API Flow

Smart RBW / VBW Cont...

Reference Validation

Protocol: Ethernet TCP/IP (Port 5000)

Algorithm: Real-Time Peak Search

Noise Floor: Dynamic -70 dBm FL...

Validation: Reference Benchmarked

Dedicated C# desktop automation platform and custom GUI interfacing natively with a Real-Time Spectrum Analyzer at ASELSAN REHİS, executing automated high-speed RF spectral signal monitoring and carrier tracking.

- ▶ **Native API Integration:** Interfaced natively via manufacturer C/C++ API over Ethernet TCP/IPv4 (Port 5000) with robust IP parsing and persistent hardware session control.
- ▶ **Parameter Management:** Implemented dynamic frequency range configuration (Start/Stop, Center, Span) with automated Resolution Bandwidth (RBW) and Video Bandwidth (VBW) tuning.
- ▶ **Proprietary Peak Search:** Designed real-time peak detection isolating carrier frequencies above calibrated noise floor (-70 dBm) to separate true signals from ambient RF noise.
- ▶ **Laboratory Benchmarking:** Conducted comparative validation against manufacturer reference software, verifying measurement accuracy in defense testing environments.

C# .NET

Spectrum Analyzer API

Ethernet TCP/IP

Peak Search Algorithm

Spectral Monitoring

RF Sniffing

TELECOM & AUTONOMOUS TELEMETRY PROJECTS

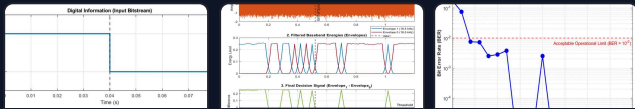
DIGITAL COMMUNICATIONS & DSP RESEARCH

Ultrasonic Ticketing System: Seamless & Offline Access Control

Near-Ultrasound Acoustic Communications · MATLAB DSP · 2026



Near-Ultrasound Acoustic Communication System Architecture



BFSK Modulated Carrier

Quadrature Mixer & DSP

BER / PSR Waterfall

Acoustic Band: 18 kHz – 22 kHz

Modulation: BFSK + Barker Sync

Channel: Two-Ray Multipath & D...

Platform: Full MATLAB Pipeline

Near-ultrasound (18–22 kHz) acoustic communication protocol enabling offline, zero-infrastructure access control without cellular data, internet, Bluetooth, or NFC, using standard commercial audio transducers.

- **Acoustic Transducers:** Utilized commercial smartphone speakers and microphones to transmit encrypted access tokens completely inaudible to human ears.
- **Digital Modulation:** Developed full MATLAB DSP pipeline featuring bit-level token synthesis, Barker code frame synchronization, and Binary Frequency Shift Keying (BFSK).
- **Channel Simulation:** Modeled multipath propagation (Two-Ray Model), ambient acoustic interference, transducer frequency limits, and motion-induced Doppler shifts.
- **DSP Receiver & Validation:** Built quadrature mixer receiver with envelope detection, parity verification, and comprehensive BER/PSR waterfall performance analysis.

MATLAB

Digital Communications

BFSK Modulation

Acoustic Channel Modeling

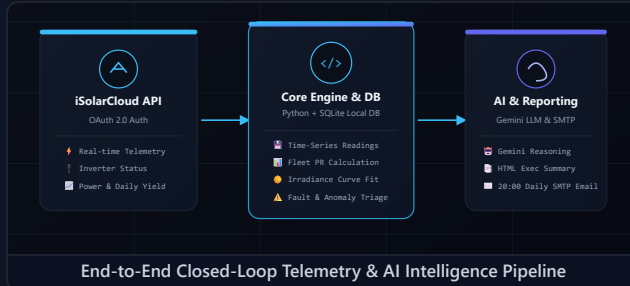
DSP

Barker Code

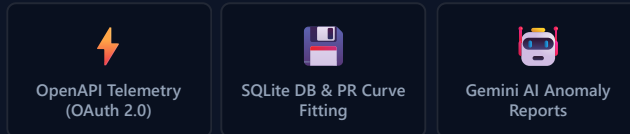
AUTONOMOUS TELEMETRY & AI SYSTEMS

SolarOPA: Autonomous AI Solar Plant Monitoring & Telemetry

Commercial PV Fleet Intelligence · Python & Gemini AI · 2026



End-to-End Closed-Loop Telemetry & AI Intelligence Pipeline



Telemetry: OpenAPI (OAuth 2.0)

Storage: SQLite Time-Series DB

Analytics: Scientific Fleet PR Curve

AI Engine: Gemini LLM (Daily SMT...

Autonomous closed-loop telemetry and operational intelligence platform for commercial solar power plants (GES) utilizing Sungrow iSolarCloud infrastructure, automating multi-inverter monitoring and anomaly diagnosis.

- **OpenAPI Telemetry:** Connected to Sungrow OpenAPI via OAuth 2.0 token management, streaming real-time multi-inverter generation, DC/AC voltages, and grid status.
- **Local Database Architecture:** Architected local SQLite time-series database storing minute-by-minute telemetry, inverter mappings, and grid fault histories.
- **Fleet PR Modeling:** Computed scientific Fleet Performance Ratio (PR) modeled against dynamic solar irradiance curves while filtering out plant anomalies.
- **Gemini AI Integration:** Prompt-engineered Google Gemini AI to analyze multi-point telemetry, diagnose anomalies, and compile executive HTML reports delivered daily at 20:00 via SMTP email.

Python

REST API

OAuth 2.0

SQLite DB

Google Gemini AI

Solar Telemetry

PROFESSIONAL ENGINEERING EXPERIENCE

Intern Engineer (RF & Microwave Systems / Test Automation) · ASELSAN REHİS Test Centers Directorate

06/2026 – 08/2026

Ankara, Turkey · Electronic Warfare & Radar Test Centers Directorate · Anechoic Chamber & Microwave Facilities

- **Planar Near-Field Antenna Scanning Automation:** Developed automated Cartesian X-Y motion control routines and C# .NET Windows Forms GUIs for anechoic chamber antenna measurements.
- **VNA 2-Port Calibration & Reference Extension:** Performed full 2-port SOLT calibrations on Vector Network Analyzers, extending measurement reference planes to test cable ends.
- **RF Power Amplifier Characterization:** Measured amplifier gain and executed P1dB 1-dB compression point tests to evaluate saturation behavior and maximum safe operating power.
- **Near-Field Chamber & NRL Arch Testing:** Conducted near-field anechoic chamber measurements and evaluated radar-absorbent material (RAM) reflectivity using NRL Arch testing.
- **Real-Time Spectrum Analyzer API & Signal Sniffing:** Integrated C/C++ API communication over Ethernet TCP/IPv4, implemented automated RBW tuning and a real-time peak detection algorithm (-70 dBm).
- **Hardware Interfacing & Safety Interlocks:** Configured Arduino Uno + CNC Shield V3 with GRBL firmware, driving stepper motors with 4x NC limit switch fail-safe cutoffs.

Intern Engineer · OPA Mühendislik

07/2025 – 08/2025 & 08/2026 – 09/2026

İzmir, Turkey · High Voltage Facilities & Solar PV Engineering

- **Solar PV Electrical Engineering:** Prepared AutoCAD single-line diagrams, electrical layout projects, and production yield simulations using PVsyst and ETAP.
- **Testing & Safety Inspections:** Conducted earth grounding resistance measurements and residual current device (RCD) tripping tests under High Voltage Operating Responsibility.
- **Reactive Power Factor Compensation:** Designed compensation panels, sized capacitor banks, and tuned reactive power relays to maintain zero reactive penalty.
- **EV Charging Infrastructure:** Calculated cable sizing, voltage drop limits, and electrical panel protections for high-power commercial EV chargers.

EDUCATION

Yaşar University – İzmir

09/2022 – Present

B.Sc. in Electrical and Electronics Engineering · 100% Merit Scholarship · GPA: 3.73 / 4.00 (High Honor Standing)

- **Relevant Coursework:** Antennas and Propagation (EEE 4340), Electromagnetic Field Theory, Microwave & RF Measurement, Digital Communications, Signals and Systems.

Cem Bakioğlu Anatolian High School

09/2018 – 06/2022

High School Education · Salutatorian (Ranked 2nd in School) · Graduation Grade: 96.89 / 100

RF & DEFENSE ENGINEERING TECHNICAL COMPETENCIES

RF & Microwave Testing

- VNA Full 2-Port SOLT Calibration
- Reference Plane Shifting to Cables
- Amplifier P1dB Compression Tests
- NRL Arch Radar Absorber Testing
- Anechoic Chamber Measurements

Motion & Embedded Systems

- Planar Cartesian X-Y Motion Control
- Arduino Uno & CNC Shield V3
- GRBL Firmware & G-Code Automation
- 4x NC Limit Switch Safety Loops
- Stepper Motor Driver Tuning

Instrumentation & Software

- C# / .NET Windows Forms GUIs
- Ethernet TCP/IPv4 Socket Streaming
- Real-Time Peak Search (-70 dBm)
- Multi-Threaded BackgroundWorkers
- MATLAB Signal Processing & DSP

CERTIFICATIONS & CREDENTIALS

- National Technology Academy:** Chip Design (Çip Tasarımı) Specialization Certificate (Code: 35867118094667) · Digital ASIC & Semiconductor Design
- ASELSAN:** "a Yetenek" Engineering Internship Program Certificate of Completion (2026) · Defense RF Systems
- ROKETSAN:** LEVEL UP AI – Smart Object Detection and Tracking with YOLO (2026) · Edge AI
- ROKETSAN:** LEVEL UP AI – Autonomous AI Agent Development & Document Assistants (2026)
- TUSAŞ (TAI):** LIFT UP Industry-Oriented Capstone Projects Program (2025) · Defense Capstone
- Savunma Sanayii Akademisi (SSB):** Milli Yetkinlik Hamlesi Competency & Career Summit (2026)
- MathWorks:** Core Signal Processing Techniques in MATLAB (Spectral Analysis, Filter Design)
- MathWorks:** Simulink Fundamentals (Model-Based Simulation) & Simscape Physical Systems
- MathWorks:** Signal Processing Onramp & MATLAB Onramp (2025)
- TÜBİTAK:** 4006 Science and Society Support Program Project Lead

LANGUAGES & INDUSTRY STATUS

- Turkish:** Native Proficiency
- English:** Professional Working Proficiency (B2 Upper-Intermediate, SFL Certified)
- Driving License:** Class B Active Driver
- Military Service:** Postponed until 31/12/2031
- Safety Credentials:** ASELSAN Occupational Health & Safety (İSG) Certified
- Internship Availability:** 2027 Spring Semester (01.02.2027 – 22.05.2027), minimum 3 full days/week on-site

PROFESSIONAL REFERENCES

Prof. Dr. Mustafa SEÇMEN

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[Repositories ↗](#)

Live Portfolio: efedemirerr.github.io

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